

Solutions to JEE Advanced - 2 | JEE 2022 | Paper - 2

PHYSICS

- 1.(4) Moment of inertia of the cylinder about the central axis,

$$I = \frac{1}{2}mR^2 = \frac{1}{2}(12.5)(0.2)^2 = 0.25 \text{ kg-m}^2$$

Now, $\tau = I\alpha = I \frac{d\omega}{dt}$

$$\Rightarrow \omega = \frac{1}{I} \int_0^{10} \tau dt = \frac{1}{I} (\text{Area under torque-time graph}) = \frac{1}{0.25} \left(\frac{1}{2} (0.2)(10) \right) = 4 \text{ rad/s}$$

- 2.(8) Let the acceleration of the blocks be
- a_1
- ,
- a_2
- and
- a_3
- respectively, all assumed upwards

Let the mass of A be m

Then, $2T - mg = ma_1$

$$T - 3g = 3a_2; \quad T - 2g = 2a_3$$

Therefore, $a_1 = \frac{2T}{m} - g; \quad a_2 = \frac{T}{3} - g; \quad a_3 = \frac{T}{2} - g$

But we know that $a_2 + a_3 = -2a_1$

$$\Rightarrow \left(\frac{T}{3} - g \right) + \left(\frac{T}{2} - g \right) = -2 \left(\frac{2T}{m} - g \right) \Rightarrow T = \frac{4g}{\frac{5}{6} + \frac{4}{m}} = \frac{24mg}{(5m + 24)}$$

But, for the block B to accelerate upwards, $a_2 > 0$

$$\Rightarrow \frac{T}{3} - g > 0 \Rightarrow T > 3g \Rightarrow \frac{24mg}{(5m + 24)} > 3g \Rightarrow m > 8$$

Alternative solution

It is quite intuitive that if the block A is heavier, the block B will accelerate upwards, and if A is lighter, B will accelerate downwards. Thus, there exists a value of the mass of A for which B remains at rest (while A and C accelerate), and this is the value that we need to find. We can find it by replacing $a_2 = 0$ in the equations.

- 3.(6) Let the density of the wooden cube be
- ρ

Initially, $(10^3)(0.6V)(10) = (\rho)(V)(10) \Rightarrow \rho = 600 \text{ kg/m}^3$

After metal block is placed, $(10^3)(V)(10) = (600)(V)(10) + (0.4)(10) \Rightarrow V = 10^{-3} \text{ m}^3$

So, mass of the cube, $m = (600)(10^{-3}) = 600 \text{ g}$

- 4.(7) Frequency of horn of car
- B
- as observed by the driver in
- A
- ,

$$f = f_0 \left(\frac{336 - (-18)}{336 - (-12)} \right) = \left(\frac{354}{348} \right) f_0 = \left(\frac{59}{58} \right) f_0$$

The observed frequency of his own horn is the same as the emitted frequency.

Hence, beat frequency,

$$f_B = f - f_0 = \frac{f_0}{58}$$

5.(8) Maximum speed, $v_M = 6(5) = 30 \text{ m/s}$

Therefore, $30 + (-2)(T - 5) = 0 \Rightarrow T = 20 \text{ s}$

So, the total time for which the car moved is 20 s.

Distance travelled between $t = 0$ and $t = 5$,

$$D_1 = \frac{1}{2}(6)(5)^2 = 75 \text{ m}$$

Distance travelled between $t = 5$ and $t = 20$,

$$D_2 = 30(15) + \frac{1}{2}(-2)(15)^2 = 225 \text{ m}$$

Since $D_1 < D_2$, we can be sure that the instant $t = T_0$ lies between $t = 5$ and $t = 20$

So, equating the distance travelled between $t = 0$ and $t = T_0$ to the distance travelled between $t = T_0$ and $t = T$,

$$\begin{aligned} 75 + 30(T_0 - 5) + \frac{1}{2}(-2)(T_0 - 5)^2 &= (30 + (-2)(T_0 - 5))(20 - T_0) + \frac{1}{2}(-2)(20 - T_0)^2 \\ \Rightarrow 30T_0 - 75 - (T_0 - 5)^2 &= 2(20 - T_0)^2 - (20 - T_0)^2 \Rightarrow 30T_0 - 75 = (20 - T_0)^2 + (T_0 - 5)^2 \\ \Rightarrow T_0^2 - 40T_0 + 250 &= 0 \Rightarrow T_0 = 20 - 5\sqrt{6} \end{aligned}$$

So, to the nearest integer, $T_0 = 8 \text{ s}$

6.(5) $W_{\text{cycle}} = W_{AB} + W_{BC} + W_{CA}$

Now, $W_{AB} = \text{Area under P-V curve} = 0$

$$\begin{aligned} W_{BC} &= -\Delta U_{BC} = -nC_V(T_C - T_B) = -2nR(T_C - T_B) \\ &= -2(P_C V_C - P_B V_B) = -2(4P_0 V_0 - 8P_0 V_0) = 8P_0 V_0 \end{aligned}$$

$$W_{CA} = \text{Area under P-V curve} = -3P_0 V_0$$

So, $W_{\text{cycle}} = 5P_0 V_0$

7.(ABD) The time period of the SHM does not depend on how the SHM is started, it only depends on the mass and the spring constant

So, the time period, $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{1}{400}} = \frac{\pi}{10} \text{ s}$

Also, the angular frequency, $\omega = \frac{2\pi}{T} = 20 \text{ rad/s}$

Let the amplitude of the resulting SHM be A

Now, since the equilibrium position is the mean position of the SHM, the initial velocity of the block must be the maximum speed during SHM, which is $A\omega$

Therefore, $A\omega = \frac{3}{2}$

$$\Rightarrow A = \frac{3}{40} = 0.075 \text{ m} = 7.5 \text{ cm}$$

Since the initial position was the mean position, there must be an extreme position above it. Therefore, the block **will** reach a position above its initial position

And the velocity of the block is never greater than the initial velocity, since the velocity is maximum at the mean position

Alternative solution

Let the elongation in the spring when the block is in equilibrium be x

By balancing forces on the block, we get $x = \frac{mg}{k} = 0.025 \text{ m}$

The equilibrium position is the mean position of the SHM.

To find the extreme position, find the position where the block comes to rest.

Let us assume that the block moves down by a distance y before coming to rest.

Taking gravitational potential energy to be zero at this final point and conserving energy,

Initial KE + Initial GPE + Initial EPE = Final KE + Final GPE + Final EPE

$$\frac{1}{2}(1)\left(\frac{3}{2}\right)^2 + (1)(10)y + \frac{1}{2}(400)(0.025)^2 = 0 + 0 + \frac{1}{2}(400)(0.025 + y)^2$$

$$\Rightarrow y = 0.075 \text{ m} = 7.5 \text{ cm}$$

The amplitude of the SHM is the separation between its mean position and its extreme position. So, the amplitude is y itself.

Since the initial position was the mean position, there must be an extreme position above it. And the velocity of the block is never greater than the initial velocity.

8.(AC) Heat current = $\frac{\text{Temperature difference}}{\text{Total thermal resistance}}$

Thermal resistance of the first rod, $R_1 = \frac{l}{K_1 A} = \frac{0.16}{400(10 \times 10^{-4})} = \frac{2}{5} \text{ K/W}$

Thermal resistance of the second rod, $R_2 = \frac{l}{K_2 A} = \frac{0.16}{160(20 \times 10^{-4})} = \frac{1}{2} \text{ K/W}$

So, the heat current, $H = \frac{T_1 - T_2}{R_1 + R_2} = \frac{200 - 20}{\left(\frac{2}{5} + \frac{1}{2}\right)} = 200 \text{ W}$

Let the temperature of the junction be T

Then, since the heat current through both rods must be equal,

$$\frac{200 - T}{R_1} = \frac{T - 20}{R_2} \Rightarrow \frac{200 - T}{\left(\frac{2}{5}\right)} = \frac{T - 20}{\left(\frac{1}{2}\right)} \Rightarrow T = 120^\circ \text{C}$$

The temperature gradient in the first rod = $\frac{200 - 120}{16} = 5^\circ \text{C/cm}$

The temperature gradient in the second rod = $\frac{120 - 20}{16} = 6.25^\circ \text{C/cm}$

9.(ABC) The cross-section of the liquid contained in the vessel is shown in the figure.

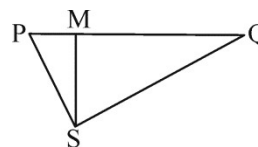
MS is perpendicular to PQ .

We are given that $SQ = L$ and $\angle MQS = \angle PSM = \theta$

Therefore, we can see that $MS = L \sin \theta$; $PM = L \sin \theta \tan \theta$; $MQ = L \cos \theta$

Horizontal component of $\vec{F}_B$,

$$(F_B)_H = (\text{Pressure at half depth}) \times (\text{Area of base projected on a vertical plane})$$



In other words, this horizontal component of force will be the same as the force that would act on the “face” MS .

$$\text{So, } (F_B)_H = \left(P_0 + \rho g \left(\frac{L \sin \theta}{2} \right) \right) \times ((L \sin \theta)(L)) = \left(P_0 L^2 \sin \theta + \frac{1}{2} \rho g L^3 \sin^2 \theta \right)$$

We can see the horizontal component of $\vec{F}_S$ will also be the same force that would act on MS . So, this component has the same magnitude as the component obtained above.

$$\text{So, } (F_S)_H = \left(P_0 + \rho g \left(\frac{L \sin \theta}{2} \right) \right) \times ((L \sin \theta)(L)) = \left(P_0 L^2 \sin \theta + \frac{1}{2} \rho g L^3 \sin^2 \theta \right)$$

Now, vertical component of $\vec{F}_B$,

$$\begin{aligned} (F_B)_V &= (\text{Weight of liquid contained in volume MSQ}) + P_0 \times (\text{Area MQ}) \\ &= \left(\rho \left(\frac{1}{2} (L \sin \theta)(L \cos \theta)(L) \right) g \right) + P_0 (L \cos \theta)(L) \\ &= \left(P_0 L^2 \cos \theta + \frac{1}{2} \rho g L^3 \sin \theta \cos \theta \right) \end{aligned}$$

Similarly, the vertical component of $\vec{F}_S$,

$$\begin{aligned} (F_S)_V &= (\text{Weight of liquid contained in volume PSM}) + P_0 \times (\text{Area PM}) \\ &= \left(\rho \left(\frac{1}{2} (L \sin \theta)(L \sin \theta \tan \theta)(L) \right) g \right) + P_0 (L \sin \theta \tan \theta)(L) \\ &= \left(P_0 L^2 \sin \theta \tan \theta + \frac{1}{2} \rho g L^3 \sin^2 \theta \tan \theta \right) \end{aligned}$$

10.(BCD) Moment of inertia of the rod about the pivot,

$$I = \frac{1}{12} mL^2 + m \left(\frac{L}{2} - \frac{L}{3} \right)^2 = \frac{1}{9} mL^2$$

So, the initial angular acceleration of the rod is given by

$$\begin{aligned} \tau &= I\alpha \\ \Rightarrow mg \left(\frac{L}{6} \right) &= \left(\frac{1}{9} mL^2 \right) \alpha \quad \Rightarrow \quad \alpha = \frac{3g}{2L} \end{aligned}$$

$$\text{So, the initial acceleration of the CM} = \alpha \left(\frac{L}{6} \right) = \frac{g}{4}$$

Let the force applied by the pivot on the rod just after it is released be F upwards

$$\text{Then, } mg - F = m \left(\frac{g}{4} \right) \quad \Rightarrow \quad F = \frac{3mg}{4}$$

The rod attains its maximum angular velocity ω when it is vertical

By conservation of energy between the horizontal and vertical positions,

$$mg \left(\frac{L}{6} \right) = \frac{1}{2} \left(\frac{1}{9} mL^2 \right) \omega^2 \quad \Rightarrow \quad \omega = \sqrt{\frac{3g}{L}}$$

And, in the vertical position, the angular acceleration of the rod is zero. Hence, the acceleration of its CM is

$$\omega^2 \left(\frac{L}{6} \right) = \frac{g}{2}$$

11.(AD) Moment of inertia of the body about the rotation axis, $I = \frac{1}{2}(2m)R^2 + mR^2 = 2mR^2$

Restoring torque, $\tau = mgR \sin \theta \approx (mgR)\theta \Rightarrow C = mgR$

Therefore, the frequency of SHM, $f = \frac{1}{2\pi} \sqrt{\frac{C}{I}} = \frac{1}{2\pi} \sqrt{\frac{g}{2R}}$

We know that the maximum KE in linear SHM is $\frac{1}{2}kA^2$. Analogously, the maximum KE in angular SHM is

$\frac{1}{2}C\theta_M^2$; So, $KE_{\max} = \frac{1}{2}mgR\theta_M^2$

12.(ABCD) Since one end of the string is $x=0$ and this point must be a node, the equation of the given stationary wave should be:

$y = A \sin(kx) \cos(\omega t)$

Comparing the given equation with this equation, we can see that

$k = \frac{5\pi}{2}$ and $\omega = 600\pi$

So, the wavelength, $\lambda = \frac{2\pi}{k} = 0.8 \text{ m}$

Since in the n^{th} harmonic, $L = n\left(\frac{\lambda}{2}\right)$ and we are given that $L = 1.2 \text{ m}$, $\Rightarrow n = 3$

So, the string vibrates in its 3rd harmonic, and it has 3 antinodes.

The antinodes are located at $x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}$ i.e at $x = 0.2, 0.6, 1.0$

Frequency of the wave, $f = \frac{\omega}{2\pi} = 300 \text{ Hz}$

Since this is the frequency of the third harmonic, the fundamental frequency is $\frac{300}{3} = 100 \text{ Hz}$

The distance between two nodes is $\frac{\lambda}{2}$, i.e. 0.4 m

13.(2) Instantaneous position of the particle, $\vec{r} = \cos(3t) \hat{i} + 2 \sin(3t) \hat{j}$

Instantaneous velocity of the particle, $\vec{v} = \frac{d}{dt}(\vec{r}) = (-3 \sin(3t)) \hat{i} + 6 \cos(3t) \hat{j}$

Instantaneous acceleration of the particle, $\vec{a} = \frac{d}{dt}(\vec{v}) = (-9 \cos(3t)) \hat{i} - 18 \sin(3t) \hat{j}$

Velocity and acceleration must be perpendicular

Therefore, $\vec{a} \cdot \vec{v} = 0$

$\Rightarrow 27 \sin(3t) \cos(3t) - 108 \sin(3t) \cos(3t) = 0$

$\Rightarrow \sin(3t) \cos(3t) = 0 \Rightarrow \sin(6t) = 0 \Rightarrow t = \frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2} \dots$

Position at $t = \frac{\pi}{6}$, $\vec{r} = \cos\left(\frac{\pi}{2}\right) \hat{i} + 2 \sin\left(\frac{\pi}{2}\right) \hat{j} = 2 \hat{j}$

- 14.(0.95) Let the initial height of mercury column be h , let the initial area of cross-section of the beaker be A . Then, then the initial volume of mercury in the beaker,

$$V = Ah \Rightarrow h = \frac{V}{A}$$

Now, after heating, let the height of mercury column be h' , let the area of cross-section of the beaker be A' and let the volume of mercury be V'

$$\text{Then, } h' = \frac{V'}{A'} = \frac{V(1 + \gamma_{Hg}(80 - 20))}{A\left(1 + \frac{2}{3}\gamma_g(80 - 20)\right)} \Rightarrow h' = h\left(\frac{1 + 60\gamma_{Hg}}{1 + 40\gamma_g}\right) = h(1 + 60\gamma_{Hg} - 40\gamma_g)$$

$$\text{So, the increase in mercury column, } \Delta h = h' - h = h(60\gamma_{Hg} - 40\gamma_g) \Rightarrow \Delta h = 0.095 \text{ cm}$$

- 15.(0.5) Let the distance of the particle from the centre of the earth when it comes to rest be x

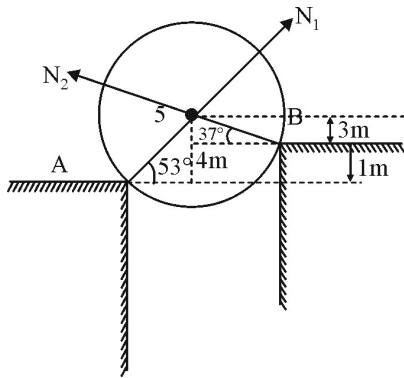
$$\text{We are given that } \frac{GM}{x^2} = \frac{9g}{16} = \frac{9GM}{16R^2} \Rightarrow x = \frac{4R}{3}$$

$$\text{Conserving energy, } K_i + U_i = K_f + U_f$$

$$\Rightarrow \frac{1}{2}mv^2 + \left(-\frac{GMm}{R}\right) = 0 + \left(-\frac{3GMm}{4R}\right) \Rightarrow v = \sqrt{\frac{GM}{2R}}$$

$$\text{Since } v_e = \sqrt{\frac{2GM}{R}}, \quad \frac{v}{v_e} = \frac{1}{2}$$

- 16.(32) Let N_1 and N_2 denote the forces acting on the sphere from the supports A and B respectively



Balancing forces,

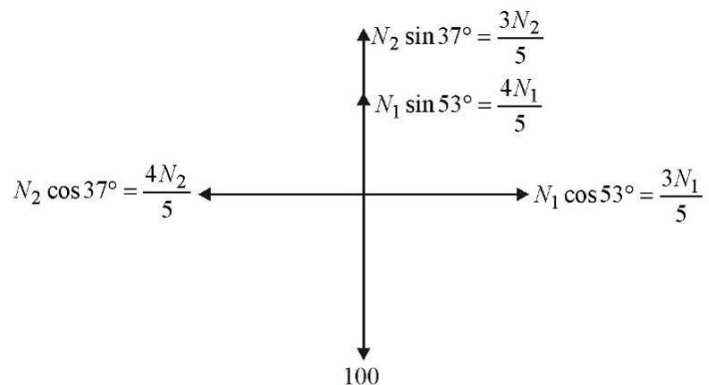
$$\frac{3N_1}{5} = \frac{4N_2}{5} \quad \dots(i)$$

$$\frac{4N_1}{5} + \frac{3N_2}{5} = 40 \quad \dots(ii)$$

Solving, we get

$$N_1 = 32 \text{ N}$$

$$N_2 = 24 \text{ N}$$



17.(50) Speed of transverse waves, $c = \sqrt{\frac{T}{\mu}}$

Now, $T = \frac{YA}{L} \Delta L$ and $\mu = \rho A$

So, $c = \sqrt{\frac{Y \Delta L}{\rho L}} = 100 \text{ m/s}$

Therefore, the fundamental frequency of the wire, $f_0 = \frac{c}{2L} = \frac{100}{2(1)} = 50 \text{ Hz}$

18.(1.35) Work done by the gas against spring force

= Increase in elastic PE

$$= \frac{1}{2} (10^6) \left((0.05)^2 - (0.02)^2 \right) = 1050 \text{ J}$$

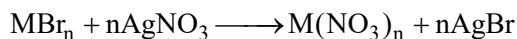
Work done by the gas against atmospheric pressure

$$= (10^5) (0.1) (0.05 - 0.02) = 300 \text{ J}$$

So, total work done = $1050 + 300 = 1350 \text{ J}$

CHEMISTRY

- 1.(3) Let oxidation state of metal ion in the metal bromide be n .

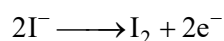


Mole of $AgNO_3$ used = $n \times$ Mole of MBr_n

$$0.025 \times \frac{60}{1000} = n \times 0.0005$$

$$n = \frac{0.025 \times 60}{1000 \times 0.0005} = 3$$

- 2.(3) $2e^- + 2H^+ + H_3AsO_4 \longrightarrow H_3AsO_3 + H_2O$



$$\text{moles of } e^- = \frac{1.5 \times 10^{22}}{6 \times 10^{23}} = \frac{1}{40} \text{ mole}$$

$$\text{moles of } I_2 = \frac{1}{80} \text{ mol}$$

$$\text{mass of } I_2 = \frac{1}{80} \times 254 = 3.175 \text{ gm}$$

- 3.(2) $Cu + 4HNO_3 \longrightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$

$$a = 1, b = 4, c = 1, d = 2, e = 2$$

- 4.(4) $n - 5 = 1 \Rightarrow n = 6$

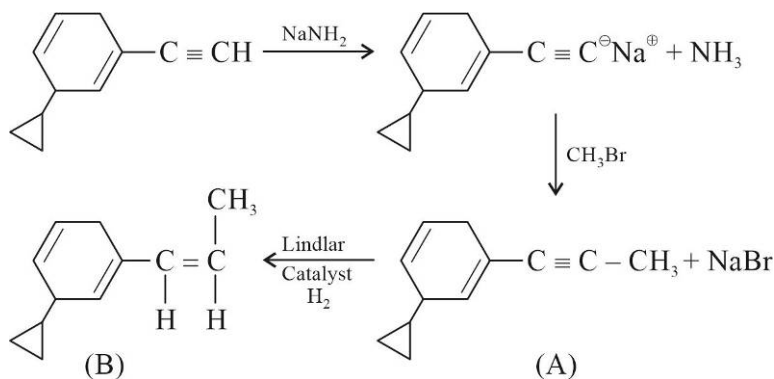
Electronic configuration of element is:

$$1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^2 3d^{10} 4p^6, 5s^2 4d^{10} 5p^6, 6s^2 4f^{10}$$

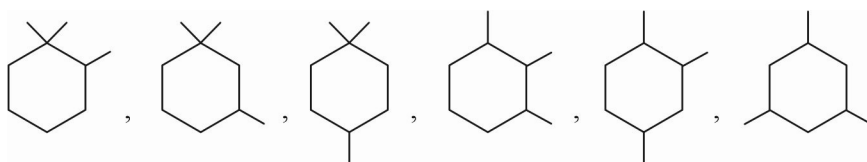
$$\text{No. of unpaired } e^- = \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow} \boxed{\uparrow} \boxed{\uparrow} \boxed{\uparrow}$$

$$= 4$$

- 5.(5)



- 6.(6)

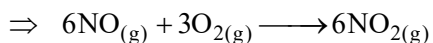
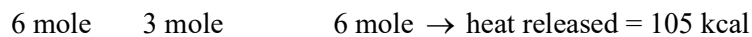
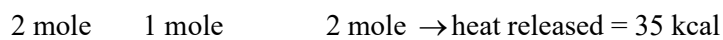
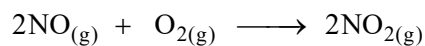


$$7.(CD) \quad \frac{P_0 V_0}{RT_0} + \frac{P_0 V_0}{RT_0} = \frac{PV_0}{RT_0} + \frac{PV_0}{R \cdot 3T_0}$$

$$\frac{2P_0}{T_0} = \frac{4P}{3T_0} \Rightarrow P = \frac{3}{2}P_0$$

$$n = \frac{P \cdot V_0}{R \cdot 3T_0} = \frac{3}{2} \frac{P_0 V_0}{R \cdot 3T_0} = \frac{P_0 V_0}{2RT_0}$$

8.(BCD)



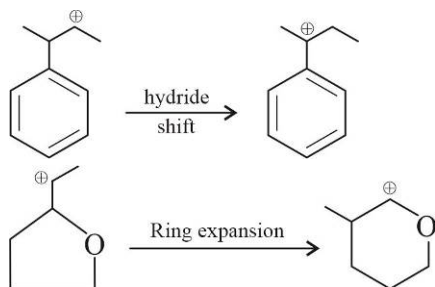
$$\Delta H = \Delta U + \Delta n_g \cdot RT$$

$$-105 = \Delta U + (-3) \times \frac{2 \times 300}{1000}$$

$$\Delta U = -105 + 1.8 = -103.2 \text{ kcal}; \quad q = \Delta U - w$$

$$-105 = -103.2 - w \Rightarrow w = 105 - 103.2 = 1.8 \text{ kcal}$$

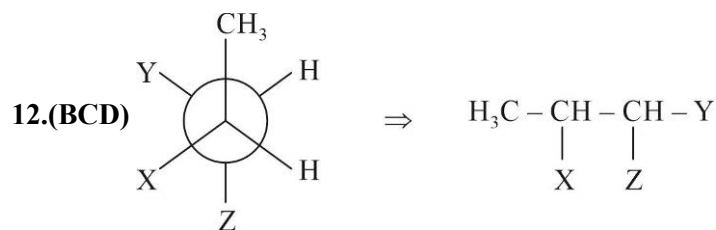
9.(AB)

10.(ABCD) The degree of dissociation of pure H_2O at $25^\circ C = 1.8 \times 10^{-7}\%$

$\therefore$ Any H^+ or OH^- ions from an external source will suppress the ionization of H_2O .

11.(ABD)

Theory based

For 2-chloro butane X can be Cl, Y can be CH_3 , Z can be H

or

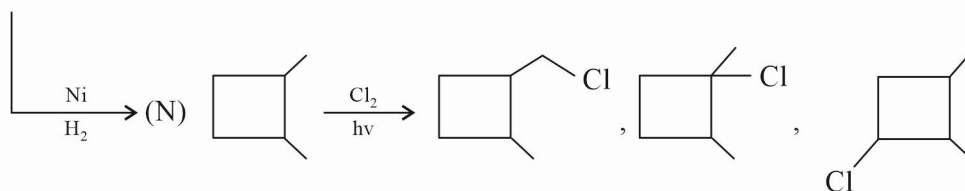
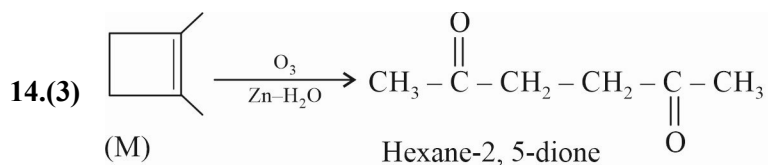
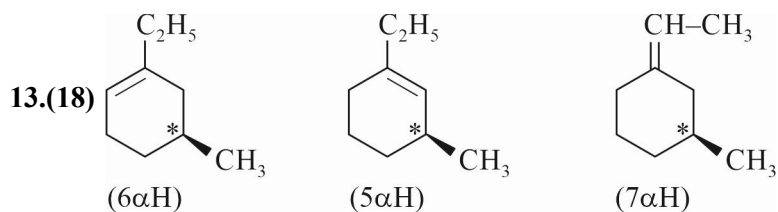
X can be Cl, Y can be H, Z can be CH_3

or

X can be H, Y can be Cl, Z can be CH_3

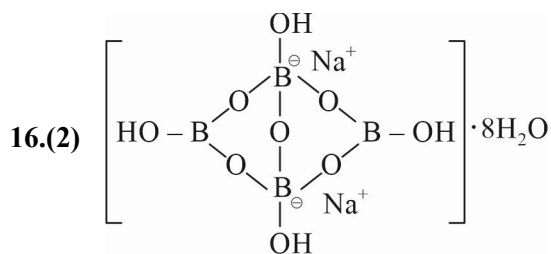
or

X can be H, Y can be CH_3 , Z can be Cl



Three isomeric monochloro products are possible.

- 15.(30) $\text{N}(\text{SiH}_3)_3$ have $p\pi - d\pi$ back bond. $a = 1$
 CH_3NCS , CO and BF_3 have $p\pi - p\pi$ back bond $b = 3$



- 17.(41) CO is neutral oxide of carbon

$$x = \frac{4}{3}, y = 1; \quad 30x + y = 41$$

- 18.(30) If x mole % of e^- are in 2^{nd} excited state then no. of moles of e^- in 2^{nd} excited state $= \frac{x}{100} \times 1$

Energy emitted by all e^- in returning from $n = 3 \rightarrow n = 1$

$$\Rightarrow \frac{x}{100} \times 1 \times 12.1 \times 96 \text{ kJ}$$

Energy emitted by all e^- in returning from $n = 2 \rightarrow n = 1$

$$\Rightarrow \frac{30}{100} \times 1 \times 10.2 \times 96 \text{ kJ}$$

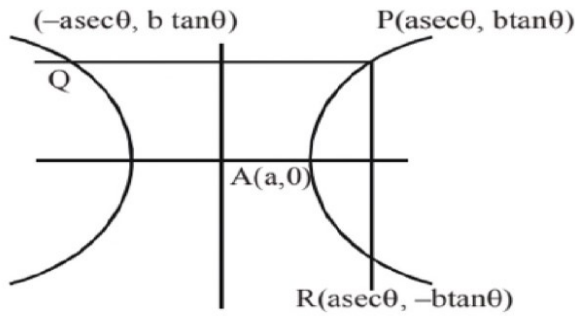
$$642.24 = \left(\frac{x}{100} \times 12.1 \times 96 \right) + \left(\frac{30}{100} \times 10.2 \times 96 \right)$$

$$642.24 = 11.616x + 293.76$$

$$x = 30$$

MATHEMATICS

1.(4)



$$(a \sec \theta)(b \tan \theta) = 2ab \tan \theta (\sec \theta - 1)$$

$$\sec \theta = 2 \sec \theta - 2$$

$$\cos \theta = \frac{1}{2}; \quad \frac{a\pi}{b} = \frac{\pi}{3} \Rightarrow a + b = 4$$

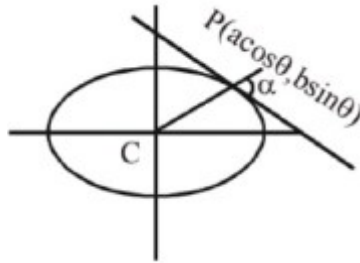
2.(5)

$$T: \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

$$m_1 = \frac{b}{a} \tan \theta$$

$$m_2 = -\frac{b}{a} \cot \theta$$

$$\tan \theta = \left| \frac{\frac{b}{a} (\tan \theta + \cot \theta)}{1 - \frac{b^2}{a^2}} \right|$$



$\Rightarrow$ minimum occurs at $\theta = 45^\circ$

3.(4)

$$\text{Let } z_1 = r_1 e^{i\theta_1} \text{ and } z_2 = r_2 e^{i\theta_2} \Rightarrow \frac{z_1 \bar{z}_2 + \bar{z}_1 z_2 + z_1 z_2 + \bar{z}_1 \bar{z}_2}{|z_1 z_2|} = 2 \cos(\theta_1 - \theta_2) + 2 \cos(\theta_1 + \theta_2)$$

4.(2)

$$\frac{a+b}{2} = \frac{5}{4} \sqrt{ab} \Rightarrow \frac{a}{b} = \frac{1}{4}$$

$$\frac{H_8 - a}{b - H_8} = 8 \times \frac{a}{b} = 8 \times \frac{1}{4} = 2$$

5.(1)

6.(8)

If a is the smallest element of such a set then $13 - a$ is the largest and the remaining elements can be chosen from $12 - 2a$ elements i.e., 2^{12-2a}

$$\text{Also, } 13 - a \geq a \Rightarrow a < 7$$

$$\therefore \text{Number of such set} = 2^{10} + 2^8 + 2^6 + \dots + 2^0 = 1365$$

$$= 3 \times 5 \times 7 \times 13$$

$\therefore$ Divisors divisible by 5 is 8.

7.(BC)

R_1	R_2	R_3	R_4	R_5	Total ways
2	1	1	1	1	${}^3C_2 \times {}^3C_1 \times {}^3C_1 = 27$
1	1	2	1	1	${}^3C_1 \times {}^3C_2 \times {}^3C_1 = 27$
1	1	1	1	2	${}^3C_1 \times {}^3C_1 \times {}^3C_2 = 27$
ways = $81 \left(\frac{6!}{2!3!} \right) = 4860$					

8.(ABC)

Let equation of direct common tangent(s) be

$$(y - 0) = m(x - 3) \Rightarrow mx - y - 3m = 0$$

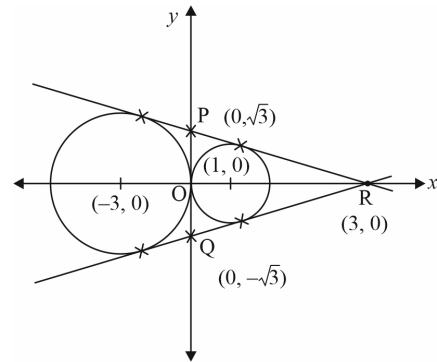
As, $p = r$

$$\Rightarrow \frac{|m(1) - 0 - 3m|}{\sqrt{1 + m^2}} = 1$$

$$\Rightarrow 4m^2 = 1 + m^2 \Rightarrow m^2 = \frac{1}{3} \Rightarrow m = \pm \frac{1}{\sqrt{3}}$$

Clearly, ΔPQR is an equilateral triangle.

Now, verify alternatives.



9.(ACD) (A) $\frac{a}{1-2a}, \frac{b}{1-2b}, \frac{c}{1-2c}$ are in H.P. $\Leftrightarrow \frac{1-2a}{a}, \frac{1-2b}{b}, \frac{1-2c}{c}$ are in A.P.

$\Leftrightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P. $\Leftrightarrow a, b, c$ are in H.P.

(B) $a - \frac{b}{2}, \frac{b}{2}, c - \frac{b}{2}$ are in G.P. $\Leftrightarrow \ln\left(a - \frac{b}{2}\right), \ln\frac{b}{2}, \ln\left(c - \frac{b}{2}\right)$ are in A.P.

(D) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P. $\Leftrightarrow e^{1/a}, e^{1/b}, e^{1/c}$ are in G.P.

10.(AD) Let $\text{Arg } z_1 = \theta$

$$\text{Arg } \bar{z}_1 = -\theta \Rightarrow \text{Arg}\left(\frac{z_1}{z_2}\right) = 2\theta = \frac{2\pi}{n} \Rightarrow \theta = \frac{\pi}{n} \dots\dots\dots(1)$$

Also $\frac{\text{Im}(z_1)}{\text{Re}(z_1)} = \sqrt{2} - 1 \Rightarrow \sqrt{2} - 1 \Rightarrow \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}(\sqrt{2} - 1) = \theta = \frac{\pi}{n} = \frac{\pi}{8} \therefore n = 8$

11.(BC) $(1+x)^m(1-x)^n; \quad 3 = {}^mC_0 \cdot {}^nC_1(-1) + {}^mC_1 \cdot {}^nC_0$

Or $3 = m - n \dots (i)$

$-6 = {}^mC_0 \cdot {}^nC_2 + {}^mC_2 \cdot {}^nC_0 - {}^mC_1 \cdot {}^nC_1$

Or $-6 = \frac{n(n-1)}{2} + \frac{m(m-1)}{2} - mn$

From (i) & (ii) $n = 9$ & $m = 12$

$$\begin{aligned}
 12.(AC) \quad & (x+a)(x+1991)+1=0 \\
 \Rightarrow & (x+a)(x+1991)=-1 \quad \Rightarrow \quad (x+a)=1 \text{ and } x+1991=-1 \\
 \Rightarrow & a=1993 \quad \text{or} \quad x+a=-1 \text{ and } x+1991=1 \quad \Rightarrow \quad a=1989
 \end{aligned}$$

13.(05142.86)

Consonants $\rightarrow b, c, d, f, g, h$ (6)

Vowels $\rightarrow a, e, i, o, u$ (5)

W:

Case I : If word begins with consonants

$$\text{Then } ({}^6C_4 \times 4!) \times ({}^5C_3 \times 3!) = 360 \times 60 = 21600$$

Case II: If word begins with vowels

$$({}^5C_4 \times 4!) \times ({}^6C_3 \times 3!) = 120 \times 120 = 14400$$

Total = 3600

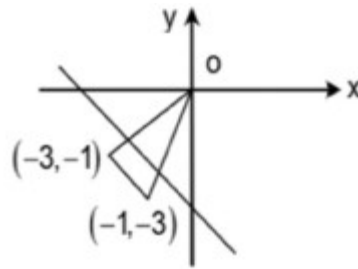
$$14.(8) \quad \text{Slope of BC} = -1 \quad \text{slope of line parallel to BC} = -\frac{a}{b}$$

$$\therefore \frac{-a}{b} = -1 \Rightarrow a = b$$

Distance of line $ax + by + 2 = 0$ from the origin

$$= \frac{2}{\sqrt{a^2 + b^2}} = \frac{1}{\sqrt{2}} \Rightarrow a^2 + b^2 = 8$$

$$\therefore a^2 = b^2 = 4 \quad \therefore a = b = 2$$



$$15.(1.27) \quad \therefore \frac{1}{4} \sin 3\theta = \frac{27}{4} \sin\left(\frac{\theta}{9}\right) - \sin^3 \theta - 3 \sin^3 \frac{\theta}{3} - 9 \sin^3\left(\frac{\theta}{9}\right)$$

$$\Rightarrow \sin 3\theta = \frac{1}{\sqrt{2}} = \sin \frac{\pi}{4} \text{ or } \sin \frac{3\pi}{4} \quad \Rightarrow \quad \theta = \frac{\pi}{12} \text{ or } \frac{\pi}{4}$$

$$\therefore \tan \frac{\pi}{12} = 2 - \sqrt{3} \text{ or } \tan \frac{\pi}{4} = 1 \quad \therefore \tan \frac{\pi}{12} + \tan \frac{\pi}{4} = 3 - \sqrt{3}$$

16.(1330)

$$\begin{aligned}
 f(x) &= (1+x)^{20} \left[\underbrace{1 + \frac{x}{1+x} + \left(\frac{x}{1+x}\right)^2 + \dots + \left(\frac{x}{1+x}\right)^{18}}_{19 \text{ terms}} \right] = (1+x)^{20} \left[\frac{1 - \left(\frac{x}{1+x}\right)^{19}}{1 - \frac{x}{1+x}} \right] \\
 &= (1+x)^{21} \left[1 - \left(\frac{x}{1+x}\right)^{19} \right] = (1+x)^{21} - \underbrace{(1+x)^2 \cdot x^{19}}_{\substack{\text{coefficient of } x^{18} \\ \text{is not possible}}}
 \end{aligned}$$

$$\therefore \text{Coefficient of } x^{18} \text{ in } f(x) = \text{coefficient of } x^{18} \text{ in } (1+x)^{21} = {}^{21}C_3 = \frac{21 \cdot 20 \cdot 19}{6} = 70 \cdot 19 = 1330$$

17.(0.2) $A(a, 0), B(0, b)$

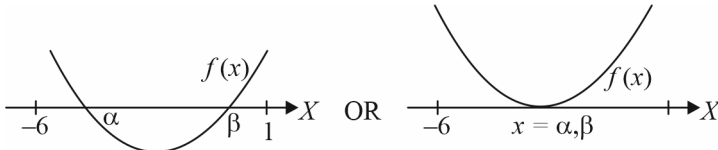
Put $y = \frac{1}{x}$

$$x^2 - ax + \frac{a}{b} = 0 \text{ and } x = \frac{1}{y}; \quad y^2 - by + \frac{b}{a} = 0$$

$$AP^2 = (x_1 - a)^2 + y_1^2 = (a - x_2 - a)^2 + (b - y_2)^2 = x_2^2 + (b - y_2)^2 = BQ^2$$

18.(12) Let $f(x) = x^2 + 2(p-3)x + 9$

Since roots lies in $(-6, 1)$, so we should have following conditions.



$$\begin{aligned} \text{(i)} \quad D &\geq 0 \Rightarrow 4(p-3)^2 - 36 \geq 0 \\ &\Rightarrow p(p-6) \geq 0 \\ &\Rightarrow p \leq 0 \text{ or } p \geq 6 \quad \dots (1) \end{aligned}$$

$$\text{(ii)} \quad f(-6) > 0 \Rightarrow p < \frac{27}{4} \quad \dots (2)$$

$$\text{(iii)} \quad f(1) > 0 \Rightarrow p > -2 \quad \dots (3)$$

$$\text{(iv)} \quad -6 < \frac{\alpha + \beta}{2} < 1 \Rightarrow 2 < p < 9 \quad \dots (4)$$

$\therefore$ From (i), (ii), (iii) & (iv),

$$\text{We get } 6 \leq p < \frac{27}{4}$$

$\therefore$ integral value of ' p ' = 6

Since $2, g_1, g_2, g_3, \dots, g_{17}, g_{18}, g_{19}, g_{20}, 6$ are in G.P.

$$\therefore g_4 g_{17} = 2 \times 6 = 12$$